

Palmer Penguins Mini Project

Anmol Sandhu

Introduction

This project examines morphological variation in penguins using the Palmer Penguins dataset. The primary objective is to assess whether the relationship between flipper length and body mass differs across species.

Reproducible research practices are essential for transparent and reliable scientific findings (Bhattacharya et al., 2025).

Research Question

Does the relationship between flipper length and body mass differ across penguin species?

Hypothesis

The relationship between flipper length and body mass differs among Adelie, Chinstrap, and Gentoo penguins. Specifically, species with longer flippers are expected to have greater body mass, but the strength of this relationship will vary among species.

Methods

Data Source

Data were obtained from the `palmerpenguins` R package and exported as a raw dataset.

Data Processing

The processed dataset was created by:

- Selecting species, flipper_length_mm, and body_mass_g
- Removing observations with missing values

The final processed dataset contains 342 observations and 3 variables.

Planned Statistical Analysis

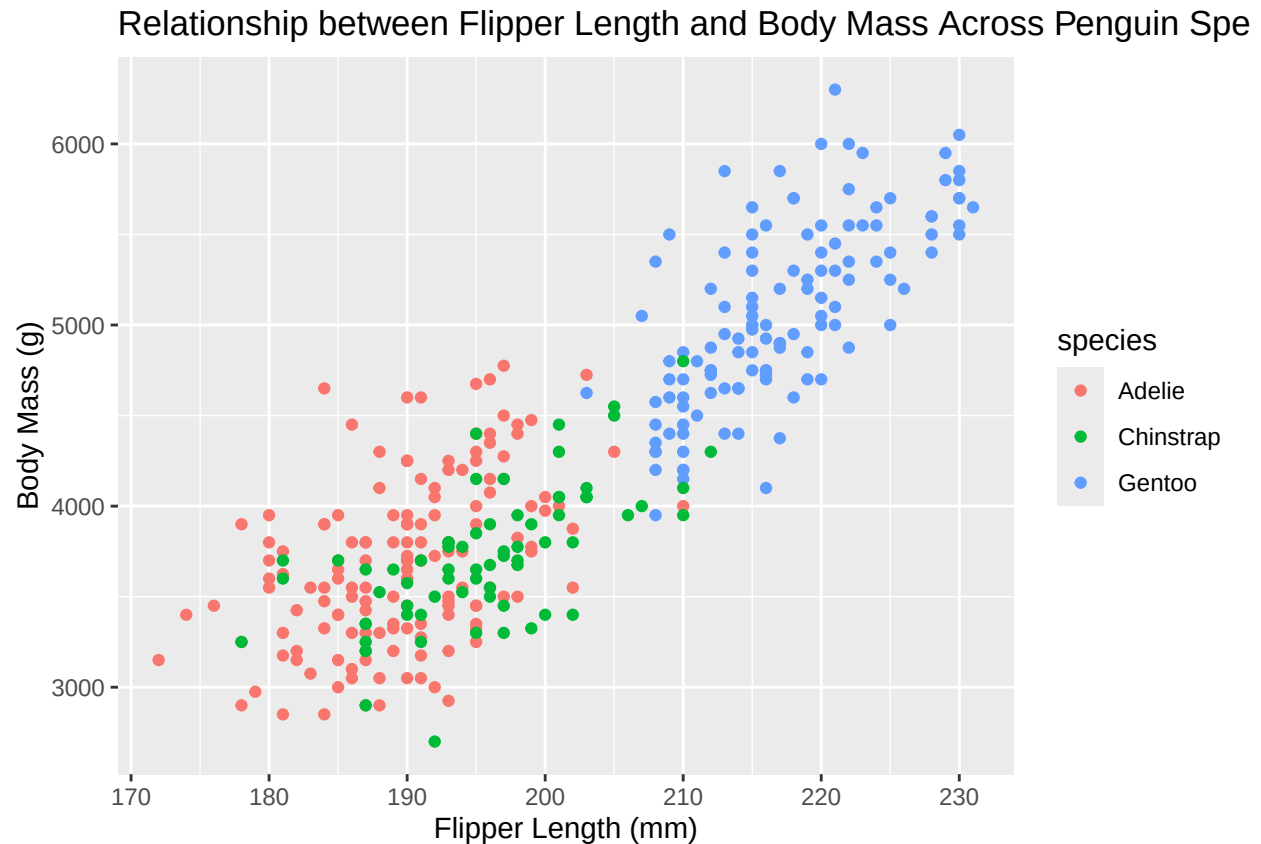


Figure 1. Relationship between flipper length and body mass across penguin species.

To evaluate whether the relationship between flipper length and body mass differs among species, a linear regression model including an interaction between flipper length and species was fitted.

```
##
## Call:
## lm(formula = body_mass_g ~ flipper_length_mm * species, data = penguins)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -911.18 -251.93  -31.77   197.82 1144.81
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -2535.837    879.468  -2.883  0.00419 **
## flipper_length_mm      32.832      4.627   7.095 7.69e-12 ***
## speciesChinstrap    -501.359   1523.459  -0.329  0.74229
## speciesGentoo     -4251.444   1427.332  -2.979  0.00311 **
## flipper_length_mm:speciesChinstrap      1.742      7.856   0.222  0.82467
## flipper_length_mm:speciesGentoo      21.791      6.941   3.139  0.00184 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 370.6 on 336 degrees of freedom
```

```
## Multiple R-squared:  0.7896, Adjusted R-squared:  0.7864
## F-statistic: 252.2 on 5 and 336 DF,  p-value: < 2.2e-16
```

Table 1. Linear regression model examining the interaction between flipper length and species on penguin body mass.

The interaction between flipper length and species was significant for Gentoo penguins ($p = 0.00184$), indicating that the relationship between flipper length and body mass differs among species.

References

Bhattacharya, S., Prusty, S., Pande, S. P., Gulhane, M., Lavate, S. H., Rakesh, N., & Veerasamy, S. (2025). Integration of multimodal imaging data with machine learning for improved diagnosis and prognosis in neuroimaging. *Frontiers in Human Neuroscience*, 19. <https://doi.org/10.3389/fnhum.2025.1552178>